Problem M2: Rapidity in an Asymmetric Collider

Question → What It Wants → Solution

Midterm Preparation Notes — v3

Reference: Key Formulas You'll Need

Rapidity Toolkit

Definition: $y = \frac{1}{2} \ln \frac{E + p_z}{E - p_z}$, equivalently: $\tanh y = \frac{p_z}{E}$

Hyperbolic relations (beam particles): $\cosh y = \gamma$, $\sinh y = \beta\gamma$, $\tanh y = \beta$

Boost property: Under a Lorentz boost along z by rapidity ϕ: $y' = y - \phi$

Beam particle approximation ($E \gg m$, moving in $+z$): $y \approx \ln \frac{2E}{m}$

Pseudorapidity: $\eta = -\ln \tan \frac{\theta}{2}$, $\eta \approx y$ when $m \ll p$

1 Part (a)

The Question (verbatim)

"Show that the rapidity of the center of mass frame y_{CM} can be calculated by finding the rapidity of the four-vector $(p_a^\mu + p_b^\mu)$. One way to do this is to write the relations to explicitly transform p_z for a and b to the center of mass, and impose the required relation between p_a and p_b there."

What It Wants

You have two objects:

1. y_{CM}: the boost rapidity that takes you from the lab to the frame where $p_{z,\text{tot}} = 0$ (the CM frame). This is a property of a *frame transformation*.
2. y_{tot}: the rapidity you compute by plugging E_{tot} and $p_{z,\text{tot}}$ into the rapidity formula. This is a property of a *four-vector*.

You must show these are equal. Do it for **general** $m_a \neq m_b$. The question tells you one approach: boost both momenta, demand $p'_z = 0$, solve.

Solution

Step 1: Define the total four-momentum.

Two particles with arbitrary masses, both moving along z:

$$P^\mu = p_a^\mu + p_b^\mu = (E_a + E_b,\ 0,\ 0,\ p_{z,a} + p_{z,b}) \equiv (E_{\text{tot}},\ 0,\ 0,\ p_{z,\text{tot}}) \tag{1}$$

No assumptions about masses or energies.

Step 2: Find y_{CM} by boosting and demanding $p'_z = 0$.

Boost the total four-momentum along z by unknown rapidity ϕ. The boosted z-component:

$$p'_{z,\text{tot}} = \gamma_\phi \left(p_{z,\text{tot}} - \beta_\phi\, E_{\text{tot}}\right) \tag{2}$$

Set to zero (the definition of the CM frame):

$$p'_{z,\text{tot}} = 0 \quad \Longrightarrow \quad \beta_\phi = \frac{p_{z,\text{tot}}}{E_{\text{tot}}} \quad \Longrightarrow \quad \tanh(\phi) = \frac{p_{z,\text{tot}}}{E_{\text{tot}}} \tag{3}$$

Convert to logarithmic form using $\tanh^{-1}(x) = \frac{1}{2}\ln\frac{1+x}{1-x}$:

$$y_{\text{CM}} = \phi = \frac{1}{2}\ln\frac{E_{\text{tot}} + p_{z,\text{tot}}}{E_{\text{tot}} - p_{z,\text{tot}}} \tag{4}$$

Step 3: Compute the rapidity of P^μ.

By the definition of rapidity applied to P^μ:

$$y_{\text{tot}} = \frac{1}{2}\ln\frac{E_{\text{tot}} + p_{z,\text{tot}}}{E_{\text{tot}} - p_{z,\text{tot}}} \tag{5}$$

Step 4: Compare.

Equations (4) and (5) are identical.

Part (a) Answer

$$\boxed{y_{\text{CM}} = y_{\text{tot}} = \frac{1}{2}\ln\frac{E_{\text{tot}} + p_{z,\text{tot}}}{E_{\text{tot}} - p_{z,\text{tot}}}} \tag{6}$$

Valid for **any** m_a, m_b, any energies. No mass assumptions used.

Why it works

The rapidity of a four-vector *is* the boost parameter of its rest frame. The CM frame *is* the rest frame of P^μ. So "$y_{\text{CM}} = y_{\text{tot}}$" is almost tautological — but that's the point. It gives a **practical recipe**: just compute E_{tot} and $p_{z,\text{tot}}$ and plug into the rapidity formula.

2 Part (b)

The Question (verbatim)

"Use this result to show that if a and b are same mass particles, that $y_{\text{CM}} = \frac{1}{2}(y_a + y_b)$."

What It Wants

1. **Specialize** part (a) to $m_a = m_b = m$.
2. **Rewrite** E_{tot} and $p_{z,\text{tot}}$ using the hyperbolic parameterization of beam four-momenta.
3. **Simplify** to show the CM rapidity is just the average of the two beam rapidities.

The key step: m cancels in the ratio because both beams have the *same* mass.

Solution

With $m_a = m_b = m$ and $p_T = 0$, the beam four-momenta are:

$$p_a^\mu = m(\cosh y_a,\ 0,\ 0,\ \sinh y_a) \tag{7}$$

$$p_b^\mu = m(\cosh y_b,\ 0,\ 0,\ \sinh y_b) \tag{8}$$

(Beam b moves in $-z$, so $y_b < 0$ and $\sinh y_b < 0$ automatically.)

Total:

$$E_{\text{tot}} = m(\cosh y_a + \cosh y_b), \qquad p_{z,\text{tot}} = m(\sinh y_a + \sinh y_b) \tag{9}$$

Plug into part (a)'s result:

$$y_{\text{CM}} = \frac{1}{2}\ln\frac{m(\cosh y_a + \sinh y_a) + m(\cosh y_b + \sinh y_b)}{m(\cosh y_a - \sinh y_a) + m(\cosh y_b - \sinh y_b)} \tag{10}$$

The mass m cancels (this is where $m_a = m_b$ is essential). Use $\cosh y + \sinh y = e^y$ and $\cosh y - \sinh y = e^{-y}$:

$$y_{\text{CM}} = \frac{1}{2}\ln\frac{e^{y_a} + e^{y_b}}{e^{-y_a} + e^{-y_b}} \tag{11}$$

Factor out $e^{(y_a+y_b)/2}$ from the numerator and $e^{-(y_a+y_b)/2}$ from the denominator:

$$\text{Num} = e^{(y_a+y_b)/2}\left(e^{(y_a-y_b)/2} + e^{-(y_a-y_b)/2}\right) = 2e^{(y_a+y_b)/2}\cosh\tfrac{y_a-y_b}{2} \tag{12}$$

$$\text{Den} = e^{-(y_a+y_b)/2}\left(e^{-(y_a-y_b)/2} + e^{(y_a-y_b)/2}\right) = 2e^{-(y_a+y_b)/2}\cosh\tfrac{y_a-y_b}{2} \tag{13}$$

The cosh factors and factors of 2 cancel:

$$y_{\text{CM}} = \frac{1}{2}\ln\frac{e^{(y_a+y_b)/2}}{e^{-(y_a+y_b)/2}} = \frac{1}{2}\ln e^{y_a+y_b} = \frac{1}{2}(y_a + y_b) \tag{14}$$

Part (b) Answer

$$\boxed{y_{\text{CM}} = \frac{1}{2}(y_a + y_b) \qquad \text{(equal masses only)}} \tag{15}$$

What happens with unequal masses

If $m_a \neq m_b$, the numerator would be $m_a e^{y_a} + m_b e^{y_b}$ and the masses would **not** cancel. You'd get a more complicated expression. The clean "average" formula is a special property of equal-mass beams.

3 Part (c)

The Question (verbatim)

"Also use the basic transformation property of y given on the slides (first derived relation/property there) to show that in the CM frame the two particles have rapidities $\pm\frac{1}{2}(y_a - y_b)$."

What It Wants

1. **Use the additive boost property** $y' = y - y_{\text{boost}}$ (or $y' = y + y_{\beta F}$ in the slide's CM→lab convention).
2. **Subtract** $y_{\text{CM}} = \frac{1}{2}(y_a + y_b)$ from each beam rapidity.
3. **Show** the results are equal and opposite: $\pm\frac{1}{2}(y_a - y_b)$.

This is a one-line calculation once you have part (b).

Solution

To go from lab to CM, boost by $y_{\text{boost}} = y_{\text{CM}} = \frac{1}{2}(y_a + y_b)$. Using $y^* = y_{\text{lab}} - y_{\text{boost}}$:

Beam a:

$$y_a^* = y_a - \frac{1}{2}(y_a + y_b) = \frac{2y_a - y_a - y_b}{2} = +\frac{1}{2}(y_a - y_b) \tag{16}$$

Beam b:

$$y_b^* = y_b - \frac{1}{2}(y_a + y_b) = \frac{2y_b - y_a - y_b}{2} = -\frac{1}{2}(y_a - y_b) \tag{17}$$

Part (c) Answer

$$\boxed{y_a^* = +\frac{1}{2}(y_a - y_b), \qquad y_b^* = -\frac{1}{2}(y_a - y_b)} \tag{18}$$

Equal and opposite, as required for the CM frame of equal-mass particles.

Sanity checks

- **Symmetric collider** ($y_a = -y_b = Y$): $y_a^* = Y$, $y_b^* = -Y$. Lab = CM. ✓
- **Fixed target** ($y_b = 0$): $y_a^* = y_a/2$, $y_b^* = -y_a/2$. Available rapidity halved. ✓
- **Signs**: $y_a > 0 > y_b$ so $y_a - y_b > 0$, giving $y_a^* > 0$, $y_b^* < 0$. ✓

4 Part (d)

The Question (verbatim, summarized)

"Suppose we wish to use sPHENIX Central Arm 'main detectors' (usual acceptance $-1 < \eta < 1$, in the CM frame with collisions positioned at the center) to study p+p physics at rapidities farther away from mid-rapidity [$y = 0$]. Thus RHIC is run with new asymmetric beam momenta of +9 GeV (the minimum 'booster' feeder proton energy to the RHIC ring) and −250 GeV (the max proton beam energy at RHIC).
Find the new CM energy ($\sqrt{s}$), y_a, and y_b. If the collisions are positioned to always happen at either end of the sPHENIX barrel, translate the resulting new η acceptance limits to what they would correspond to in the CM frame of reference.
...assume the $\Delta\eta$ size of the acceptance of sPHENIX always remains the same (= 2), and...the acceptance to go now between 0 and ±2...
make sure you justify why in this case by considering η (= y!) of detected particles on the same end of the barrel as the collision point."

What It Wants — Broken Into Tasks

1. **Compute $\sqrt{s}$**: CM energy from two beams of 250 GeV and 9 GeV protons.
2. **Compute y_a and y_b**: Rapidity of each beam. Use beam particle approximation ($E \gg m$).
3. **Compute y_{CM}**: Use part (b).
4. **Understand the geometry**: The sPHENIX barrel has a fixed physical extent covering $\Delta\eta = 2$. In symmetric mode, collisions happen at the center and coverage is $-1 < \eta < +1$. In asymmetric mode, collisions are at each *end* of the barrel, giving coverage $[0, +2]$ or $[-2, 0]$.
5. **Translate to CM frame**: Convert the lab-frame η limits to CM-frame y^* values by subtracting y_{CM}.
6. **Justify $\eta = y$**: Explain why pseudorapidity ≈ rapidity for the detected particles, so that the shift formula works.

Task 1: Compute $\sqrt{s}$

Beam energies: $E_a = 250$ GeV (in $+z$), $E_b = 9$ GeV (in $-z$). Both protons ($m_p = 0.9383$ GeV).
For $E \gg m$, the four-momenta are:

$$p_a^\mu \approx (E_a, 0, 0, +E_a), \qquad p_b^\mu \approx (E_b, 0, 0, -E_b) \tag{19}$$

$$s = (p_a + p_b)^2 = (E_a + E_b)^2 - (E_a - E_b)^2 = 4E_aE_b \tag{20}$$

$\sqrt{s}$

$$\boxed{\sqrt{s} = 2\sqrt{E_aE_b} = 2\sqrt{250 \times 9} = 2\sqrt{2250} \approx 94.9 \text{ GeV}} \tag{21}$$

Task 2: Compute y_a and y_b

For a beam particle moving in $+z$ with $E \gg m$:

$$y_a = \frac{1}{2}\ln\frac{E_a + p_a}{E_a - p_a} \approx \frac{1}{2}\ln\frac{2E_a}{m^2/2E_a} = \frac{1}{2}\ln\frac{4E_a^2}{m^2} = \ln\frac{2E_a}{m} \tag{22}$$

$$y_a = \ln\frac{2(250)}{0.9383} = \ln(532.9) \approx 6.28 \tag{23}$$

For beam b moving in $-z$ ($p_z = -E_b$):

$$y_b = \frac{1}{2}\ln\frac{E_b - E_b}{E_b + E_b} \approx -\ln\frac{2E_b}{m} = -\ln\frac{18}{0.9383} = -\ln(19.18) \approx -2.95 \tag{24}$$

Beam rapidities

$$\boxed{y_a \approx +6.28, \qquad y_b \approx -2.95} \tag{25}$$

Task 3: Compute y_{CM}

From part (b), with both beams being protons ($m_a = m_b$):

$$y_{\text{CM}} = \frac{1}{2}(y_a + y_b) = \frac{1}{2}(6.28 - 2.95) = \frac{1}{2}(3.33) \tag{26}$$

y_{CM}

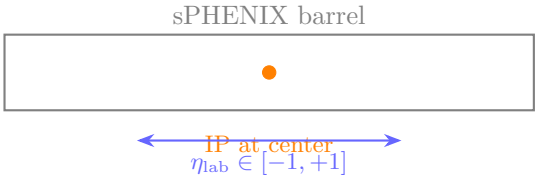
$$\boxed{y_{\text{CM}} \approx +1.66} \tag{27}$$

The CM frame moves in the $+z$ direction (chasing the 250 GeV beam).

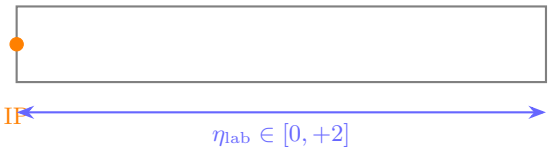
Cross-check: $y_{\text{CM}} = \frac{1}{2}\ln\frac{E_a}{E_b} = \frac{1}{2}\ln\frac{250}{9} = \frac{1}{2}(3.325) = 1.66$ ✓

Task 4: Understand the geometry

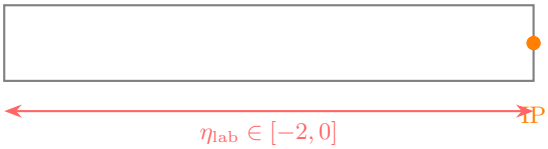
Symmetric mode:



Asym, IP at left end:



Asym, IP at right end:



Key point: the barrel is a **fixed physical object**. Its angular coverage doesn't change. What changes is where the collision happens *relative to the barrel*. When the IP moves to one end, the barrel only sees particles going in one direction from the collision point, covering $\Delta\eta = 2$ on one side of $\eta = 0$.

Task 5: Translate to CM frame

The shift formula:

$$y^*_{\mathrm{CM}} = y_{\mathrm{lab}} - y_{\mathrm{CM}} \approx \eta_{\mathrm{lab}} - y_{\mathrm{CM}} \tag{28}$$

(The last step uses $\eta \approx y$; justified in Task 6.)

IP at left end ($\eta_{\mathrm{lab}} \in [0, +2]$):

$$y^* \in [0 - 1.66,\ \ +2 - 1.66] = [-1.66,\ \ +0.34] \tag{29}$$

IP at right end ($\eta_{\mathrm{lab}} \in [-2, 0]$):

$$y^* \in [-2 - 1.66,\ \ 0 - 1.66] = [-3.66,\ \ -1.66] \tag{30}$$

Part (d) — CM-Frame Acceptance

	Lab frame η	**CM frame y^***
Symmetric (reference)	$[-1, +1]$	$[-1, +1]$
Asym, IP at left end	$[0, +2]$	$\boxed{[-1.66, +0.34]}$
Asym, IP at right end	$[-2, 0]$	$\boxed{[-3.66, -1.66]}$

Task 6: Justify $\eta = y$

The question explicitly says to justify this. The relation between rapidity and pseudorapidity for a particle with mass m, transverse momentum p_T, and rapidity y is:

$$y = \frac{1}{2} \ln \frac{\sqrt{p_T^2 \cosh^2 y + m^2} + p_T \sinh y}{\sqrt{p_T^2 \cosh^2 y + m^2} - p_T \sinh y} \tag{31}$$

In the limit $m \to 0$ (or equivalently $m \ll p$), this reduces to $y \to \eta$. For the particles detected in sPHENIX (mostly pions with $m_\pi \approx 0.14$ GeV at $p_T > 0.5$–1 GeV), the mass is a small correction and $\eta \approx y$ is an excellent approximation.

Why this matters: the additive property $y' = y - y_{\mathrm{boost}}$ is exact for *rapidity*. Pseudorapidity does **not** shift by a constant under boosts in general. But if $\eta \approx y$ for all detected particles, then we can use:

$$y^*_{\mathrm{CM}} \approx \eta_{\mathrm{lab}} - y_{\mathrm{CM}} \tag{32}$$

which is what we did in Task 5. Without $\eta \approx y$, the translation would be more complicated.

Complete Summary

Part	What It Asks	Answer	Key Step
(a)	Show $y_{\text{CM}} = y_{\text{tot}}$ for general masses	$\frac{1}{2}\ln\frac{E_{\text{tot}}+p_{z,\text{tot}}}{E_{\text{tot}}-p_{z,\text{tot}}}$	Boost, set $p'_z = 0$, compare with rapidity definition
(b)	Specialize to $m_a = m_b$, show $y_{\text{CM}} = \frac{1}{2}(y_a + y_b)$	$\frac{1}{2}(y_a + y_b)$	m cancels, use $e^y = \cosh y + \sinh y$
(c)	Find beam rapidities in CM	$\pm\frac{1}{2}(y_a - y_b)$	Subtract y_{CM} from each
(d)	RHIC numbers + translate detector acceptance	See table above	$s = 4E_aE_b$, shift η by y_{CM}, justify $\eta = y$

The Physics of Part (d)

- In symmetric 250+250 running: sPHENIX sees $y^* \in [-1, +1]$ (mid-rapidity).
- Going asymmetric (250+9) and moving the IP to the right end of the barrel: sPHENIX reaches $y^* \approx -3.7$. This is deep into the **forward region** of the 9 GeV beam.
- At forward rapidity, partons in the 250 GeV proton have very small x (gluon-dominated), while partons in the 9 GeV proton have large x. This probes the **gluon saturation** regime.
- All achieved with the same detector, just by choosing beam energies and collision point location.

Connection to PEP-II (from Thomson, Ch. 14)

This is the same trick used at SLAC's PEP-II B-factory: 9 GeV e^- on 3.1 GeV e^+ gave an asymmetric boost of $\beta\gamma = 0.56$, enough to separate B-meson decay vertices that would overlap in a symmetric collider. Different physics goal, identical mathematical principle: asymmetric beams $\Rightarrow$ CM boost $\Rightarrow$ exploit the boost to see something you otherwise couldn't.